

경북대학교 IDEC 2026 창의 회로설계 챌린지

디지털 분야 교육 Sample Flow

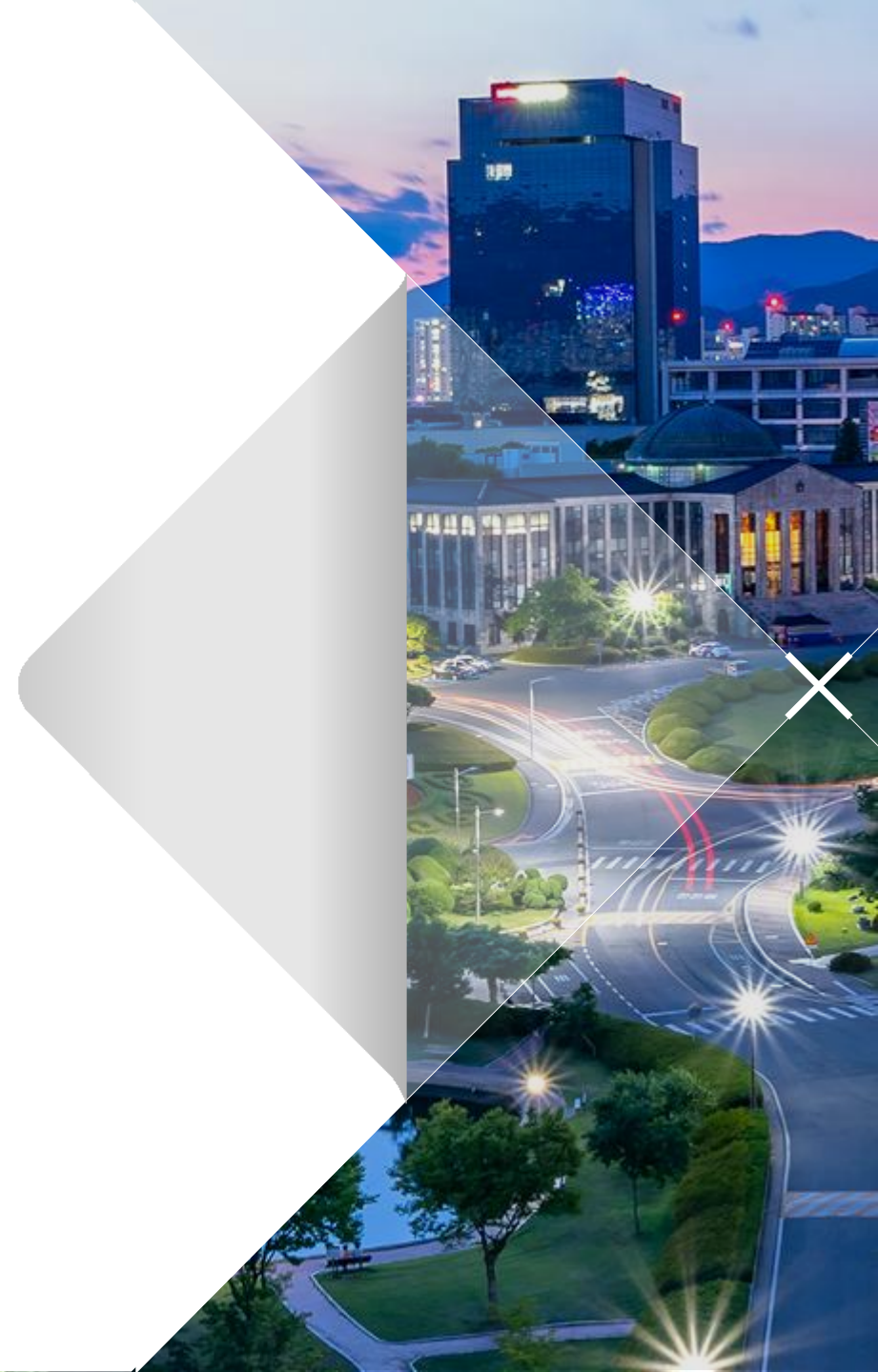
Sunmean Kim

CONTACT

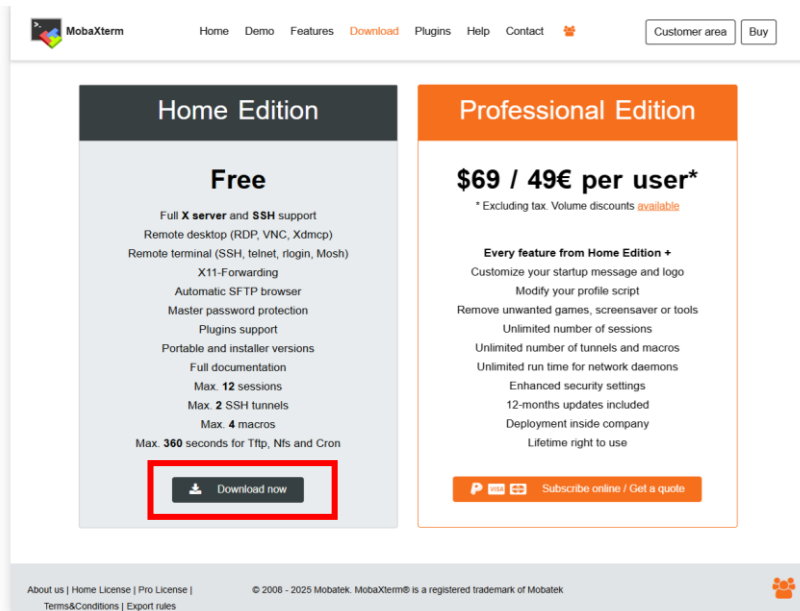
Kyungpook National University
School of Electronics Engineering
Lab. <https://ice.knu.ac.kr>
Tel. +82-53-950-5531
E-mail sunmean@knu.ac.kr



I. Ubuntu Setup



<https://mobaxterm.mobatek.net/download.html>



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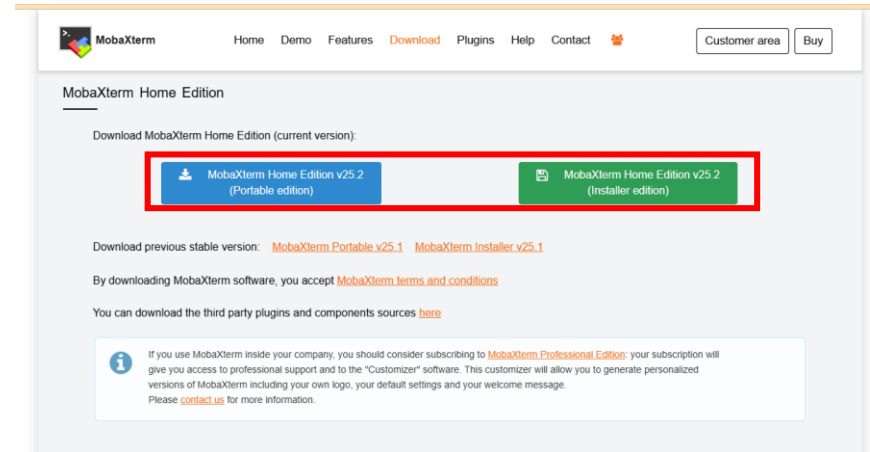
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MobaXterm Home Edition

Download MobaXterm Home Edition (current version):

[MobaXterm Home Edition v25.2 \(Portable edition\)](#) [MobaXterm Home Edition v25.2 \(Installer edition\)](#)

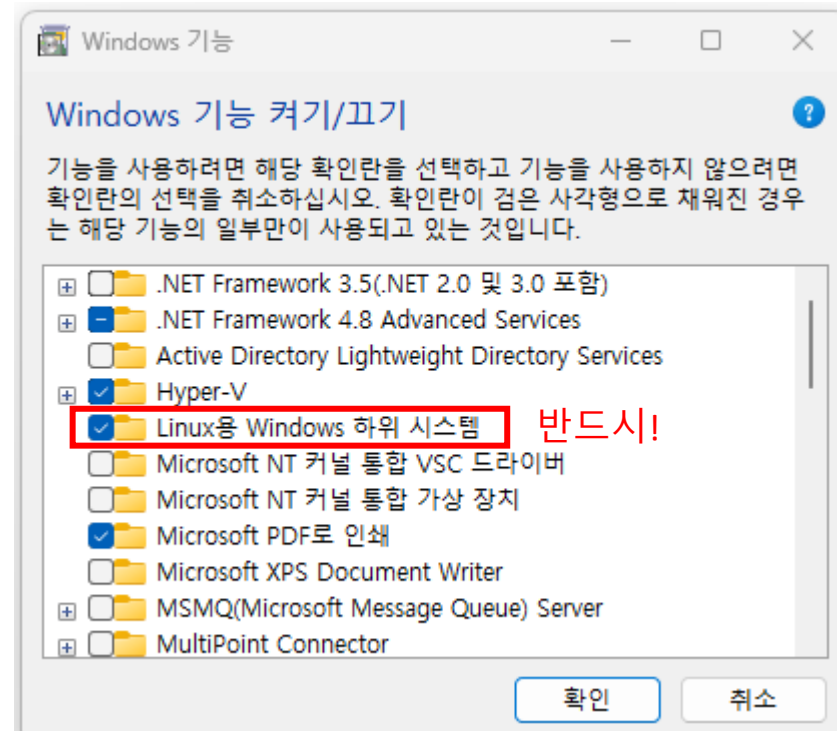
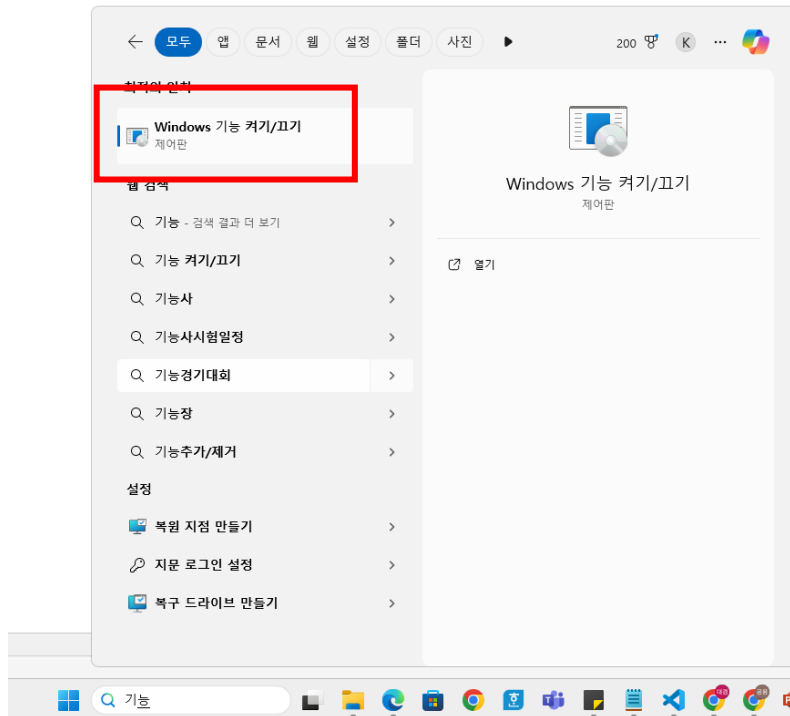
Download previous stable version: [MobaXterm Portable v25.1](#) [MobaXterm Installer v25.1](#)

By downloading MobaXterm software, you accept [MobaXterm terms and conditions](#)

You can download the third party plugins and components sources [here](#)

If you use MobaXterm inside your company, you should consider subscribing to [MobaXterm Professional Edition](#): your subscription will give you access to professional support and to the "Customizer" software. This customizer will allow you to generate personalized versions of MobaXterm including your own logo, your default settings and your welcome message. Please [contact us](#) for more information.

둘 다 괜찮음



<https://apps.microsoft.com/detail/9pn20msr04dw?hl=en-US&gl=US>

Microsoft | Store | Home | Apps | Games | Entertainment | About



Ubuntu 22.04.5 LTS

Canonical Group Limited

4.2 ★ | 493 ratings | [Developer tools](#)

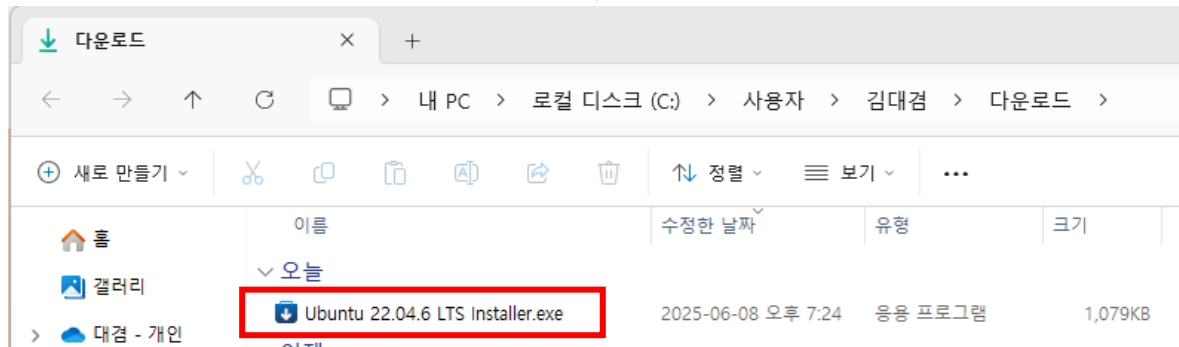


EVERYONE

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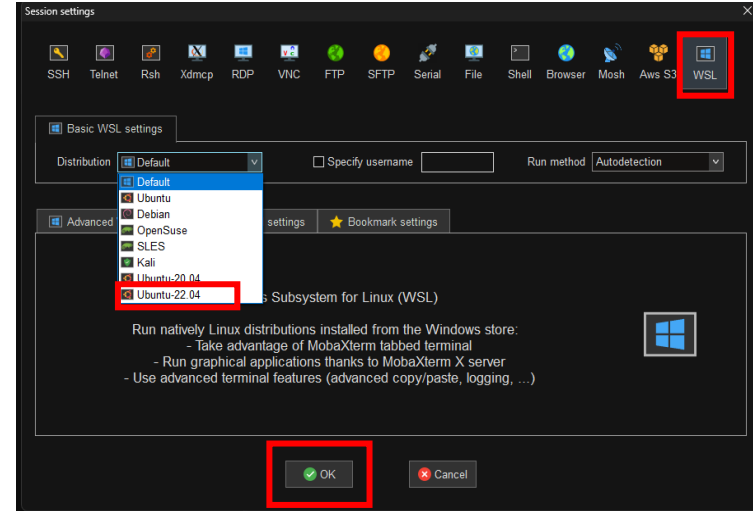
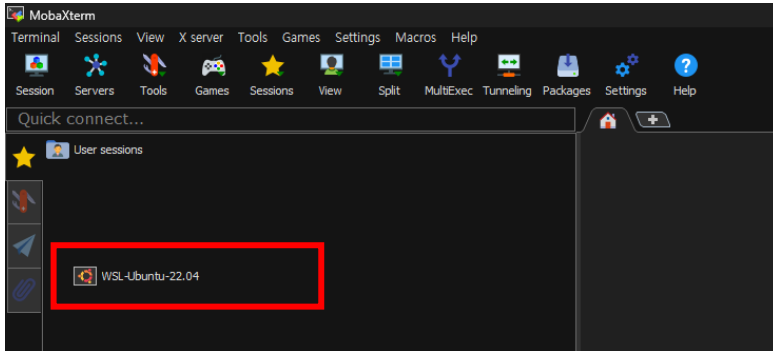
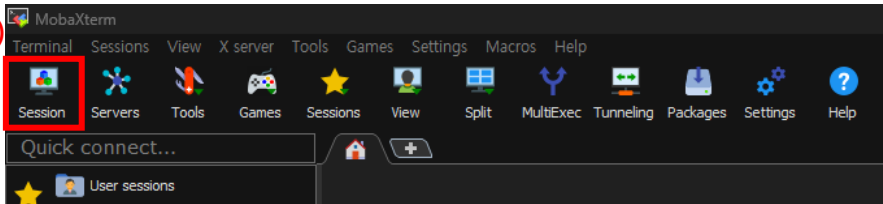
View in Store





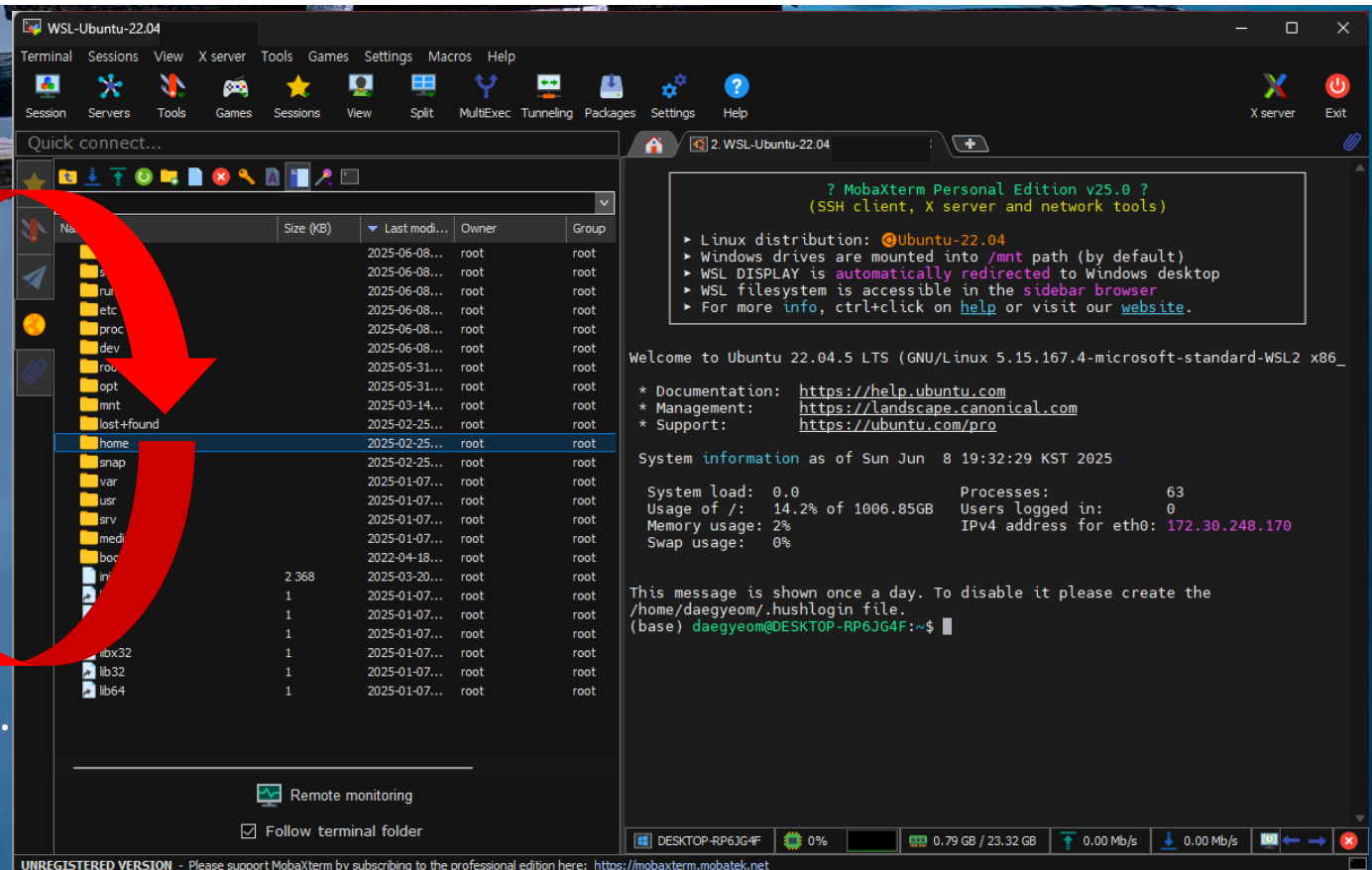
```
Installing, this may take a few minutes...  
Please create a default UNIX user account. The username does not need to match your Windows username.  
For more information visit: https://aka.ms/wslusers  
Enter new UNIX username: anythingyouwant  
New password:  
Retype new password:  
passwd: password updated successfully  
Installation successful!  
|
```

ID, password 설정(실행 후 터미널 자동 실행됨.)
설정 다 되면 ubuntu 터미널 종료 후 mobaxterm 실행

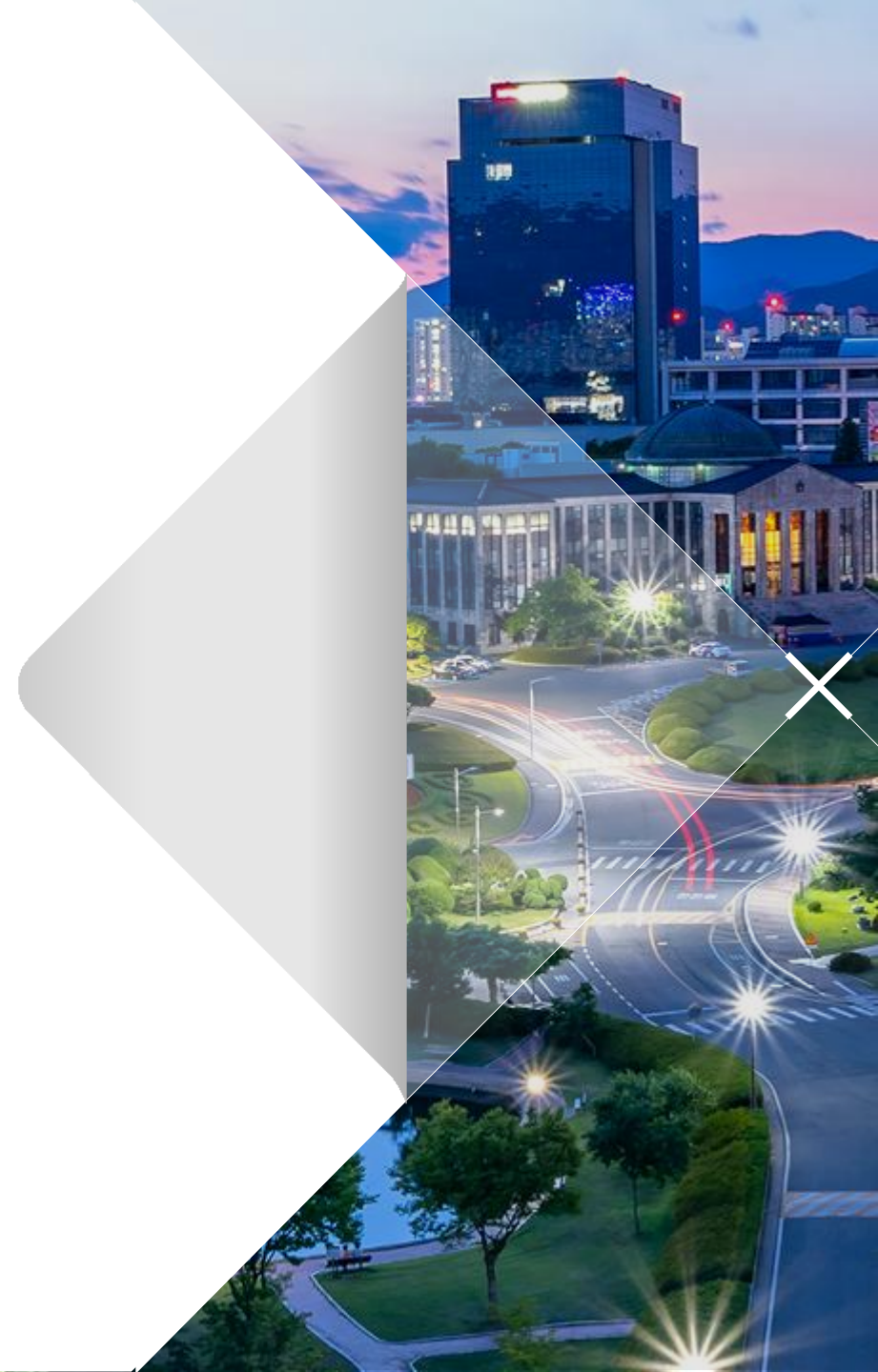




파일 끌어서 옮길 수 있음.



II. OpenROAD Setup



Build OpenROAD with local



<https://github.com/The-OpenROAD-Project/OpenROAD-flow-scripts/tree/master/docs>

<https://openroad-flow-scripts.readthedocs.io/en/latest/user/BuildLocally.html>

```
? MobaXterm Personal Edition v25.0 ?
(SSH client, X server and network tools)

> Linux distribution: Ubuntu-22.04
> Windows drives are mounted into /mnt path (by default)
> WSL DISPLAY is automatically redirected to Windows desktop
> WSL filesystem is accessible in the sidebar browser
> For more info, ctrl+click on help or visit our website.

Welcome to Ubuntu 22.04.5 LTS (GNU/Linux 5.15.167.4-microsoft-standard-WSL2 x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Sun Jun  8 19:32:29 KST 2025

System load:  0.0          Processes:    63
Usage of /:   14.2% of 1006.85GB   Users logged in:  0
Memory usage: 2%           IPv4 address for eth0: 172.30.248.170
Swap usage:   0%

This message is shown once a day. To disable it please create the
/home/daegyeom/.hushlogin file.
(base) daegyeom@DESKTOP-RP6JG4F:~$
```

붙여넣기 : 우클릭

```
git clone --recursive https://github.com/The-OpenROAD-Project/OpenROAD-flow-scripts.git
```

```
cd OpenROAD-flow-scripts
sudo ./setup.sh
cd docs
sudo apt install python3-virtualenv
virtualenv .venv
source .venv/bin/activate
pip install -r requirements.txt
make html
cd ..
```

```
sudo ./build_openroad.sh --local
```

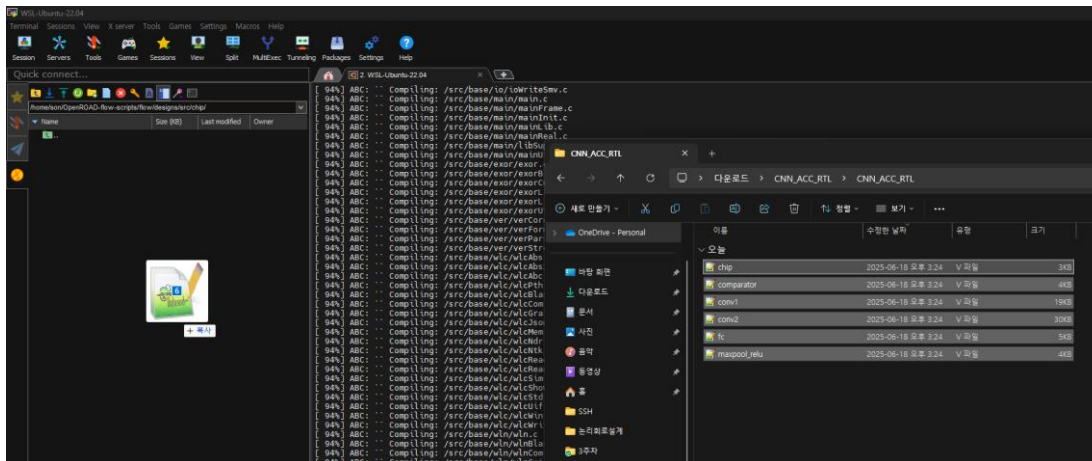
OpenROAD Project



Cloud > 2. Class > 논리회로설계 > openroad_project >					
정렬 보기 ...					
이름	상태	수정한 날짜	유형	크기	
config		2025-06-21 오후 2:54	파일 폴더		
std_cell		2025-06-21 오후 2:45	파일 폴더		
synth result		2025-06-21 오후 2:50	파일 폴더		
verilog		2025-06-21 오후 2:44	파일 폴더		

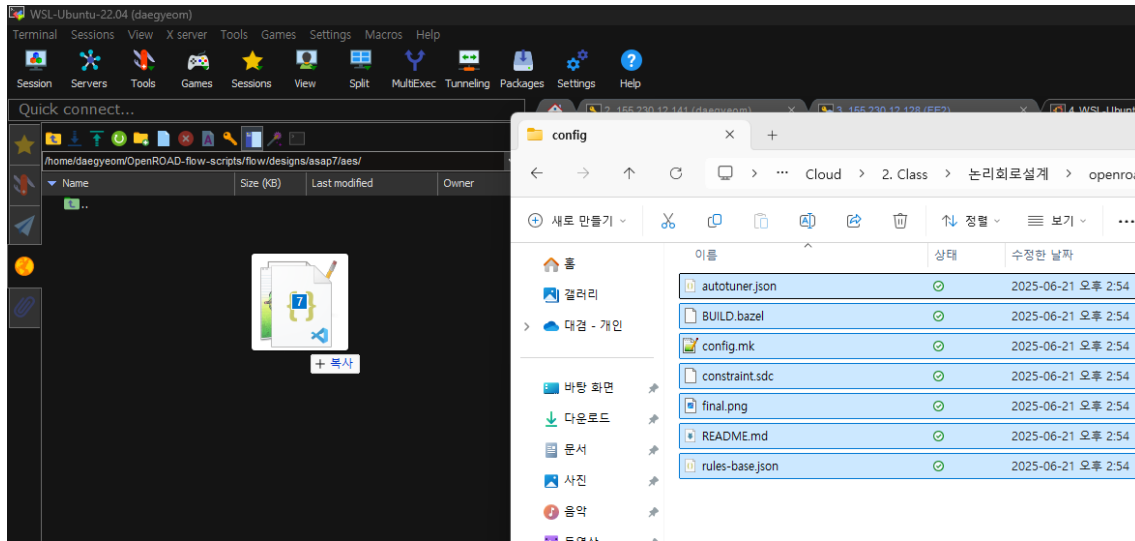
← RTL code

mkdir ~/OpenROAD-flow-scripts/flow/designs/src/chip # 디렉토리 생성 후 복붙



↶ 🖥️ > ... Cloud > 2. Class > 논리회로설계 > openroad_project >				
✂️ 📄 📁 📄 📄 🗑️ ↕️ 정렬 ▾ ≡ 보기 ▾ ...				
이름	상태	수정한 날짜	유형	크기
📁 config ← config	🔄	2025-06-21 오후 2:54	파일 폴더	
📁 std_cell	✅	2025-06-21 오후 2:45	파일 폴더	
📁 synth result	☁️	2025-06-21 오후 2:50	파일 폴더	
📁 verilog	✅	2025-06-21 오후 2:44	파일 폴더	

mkdir ~/OpenROAD-flow-scripts/flow/designs/asap7/chip # 디렉토리 생성 후 복붙



```
config.mk
~/OpenROAD-flow-scripts/flow/designs/asap7/aes

1 export PLATFORM = asap7
2
3 export DESIGN_NAME = aes_cipher_top
4 export DESIGN_NICKNAME = aes
5
6 export VERILOG_FILES = $(sort $(wildcard ./designs/src/$(DESIGN_NICKNAME)/*.v))
7 export SDC_FILE = ./designs/$(PLATFORM)/$(DESIGN_NICKNAME)/constraint.sdc
8
9 export ABC_AREA = 1
10
11 export CORE_UTILIZATION = 40
12 export CORE_ASPECT_RATIO = 1
13 export CORE_MARGIN = 2
14 export PLACE_DENSITY = 0.65
15 export TNS_END_PERCENT = 100
16 export EQUIVALENCE_CHECK ?= 1
17 export REMOVE_CELLS_FOR_EQY = TAPCELL*
```

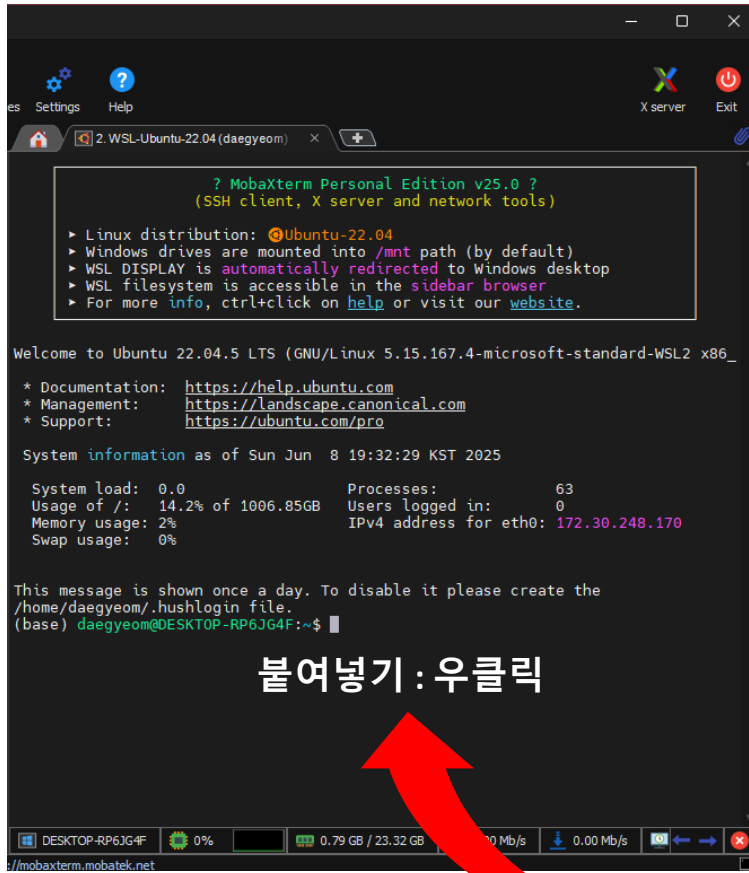
사용 platform(공정) 이름

Verilog code 위치 파일 이름

Top module name

<https://github.com/The-OpenROAD-Project/OpenROAD-flow-scripts/tree/master/docs>

<https://openroad-flow-scripts.readthedocs.io/en/latest/user/BuildLocally.html>



```
? MobaXterm Personal Edition v25.0 ?
(SSH client, X server and network tools)

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* Support: https://ubuntu.com/pro

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Usage of /:   14.2% of 1006.85GB   Users logged in:  0
Memory usage: 2%            IPv4 address for eth0: 172.30.248.170
Swap usage:   0%

This message is shown once a day. To disable it please create the
/home/daegyeom/.hushlogin file.
(base) daegyeom@DESKTOP-RP6JG4F:~$
```

붙여넣기 : 우클릭

#반드시 경로 OpenROAD-flow-scripts/에서 실행

source ./env.sh

cd flow

make DESIGN_CONFIG=./designs/asap7/chip/config.mk

<https://openroad-flow-scripts.readthedocs.io/en/latest/user/BuildWithDocker.html>

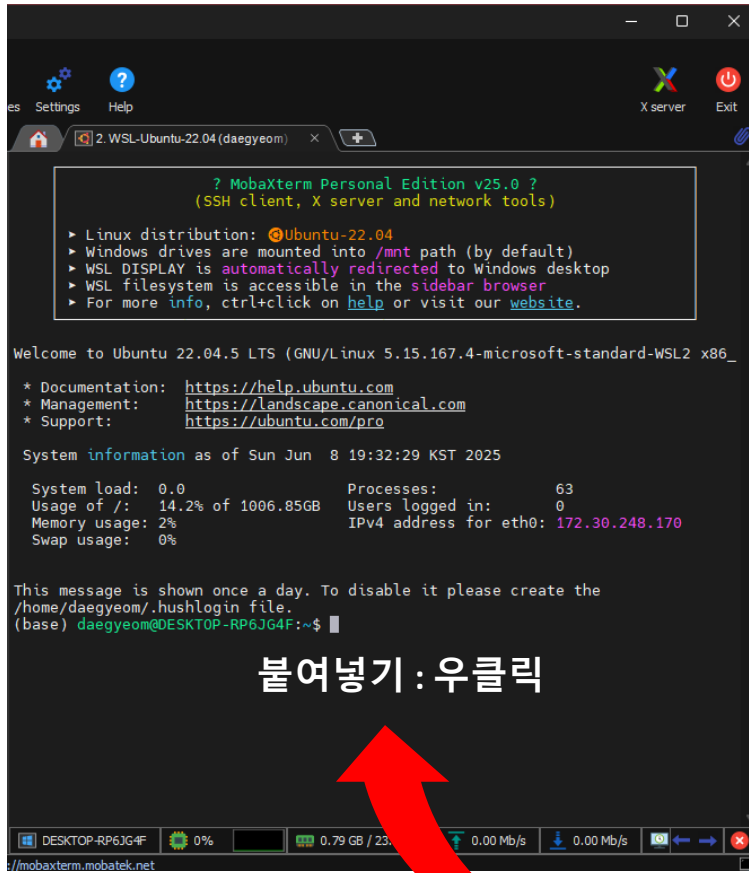
#Local 실행(이전 방법) 안될 때,
#Docker 사용 build 및 사용

Add Docker's official GPG key:

```
sudo apt-get update
sudo apt-get install ca-certificates curl
sudo curl -fsSL https://download.docker.com/linux/ubuntu/gpg -o
/etc/apt/keyrings/docker.asc
sudo chmod a+r /etc/apt/keyrings/docker.asc
```

Add the repository to Apt sources:

```
echo \
"deb [arch=$(dpkg --print-architecture) signed-
by=/etc/apt/keyrings/docker.asc]
https://download.docker.com/linux/ubuntu \
$(. /etc/os-release && echo "${UBUNTU_CODENAME:-
$VERSION_CODENAME}") stable" | \
sudo tee /etc/apt/sources.list.d/docker.list > /dev/null
sudo apt-get update
sudo apt-get install docker-ce docker-ce-cli containerd.io docker-
buildx-plugin docker-compose-plugin
```



붙여넣기 : 우클릭

Build OpenROAD with docker



<https://openroad-flow-scripts.readthedocs.io/en/latest/user/BuildWithDocker.html>

#Local 실행(이전 방법) 안될 때,
#Docker 사용 build 및 사용

git clone --recursive <https://github.com/The-OpenROAD-Project/OpenROAD-flow-scripts.git>

```
cd OpenROAD-flow-scripts
sudo ./build_openroad.sh --threads 10
```

```
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Usage of /:   14.2% of 1006.85GB   Users logged in:  0
Memory usage: 2%           IPv4 address for eth0: 172.30.248.170
Swap usage:   0%

This message is shown once a day. To disable it please create the
/home/daegyeom/.hushlogin file.
(base) daegyeom@DESKTOP-RP6JG4F:~$
```

붙여넣기 : 우클릭



```
? MobaXterm Personal Edition v25.0 ?
(SSH client, X server and network tools)

> Linux distribution: Ubuntu-22.04
> Windows drives are mounted into /mnt path (by default)
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Welcome to Ubuntu 22.04.5 LTS (GNU/Linux 5.15.167.4-microsoft-standard-WSL2 x86_64)

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Usage of /:   14.2% of 1006.85GB   Users logged in:    0
Memory usage: 2%           IPv4 address for eth0: 172.30.248.170
Swap usage:   0%

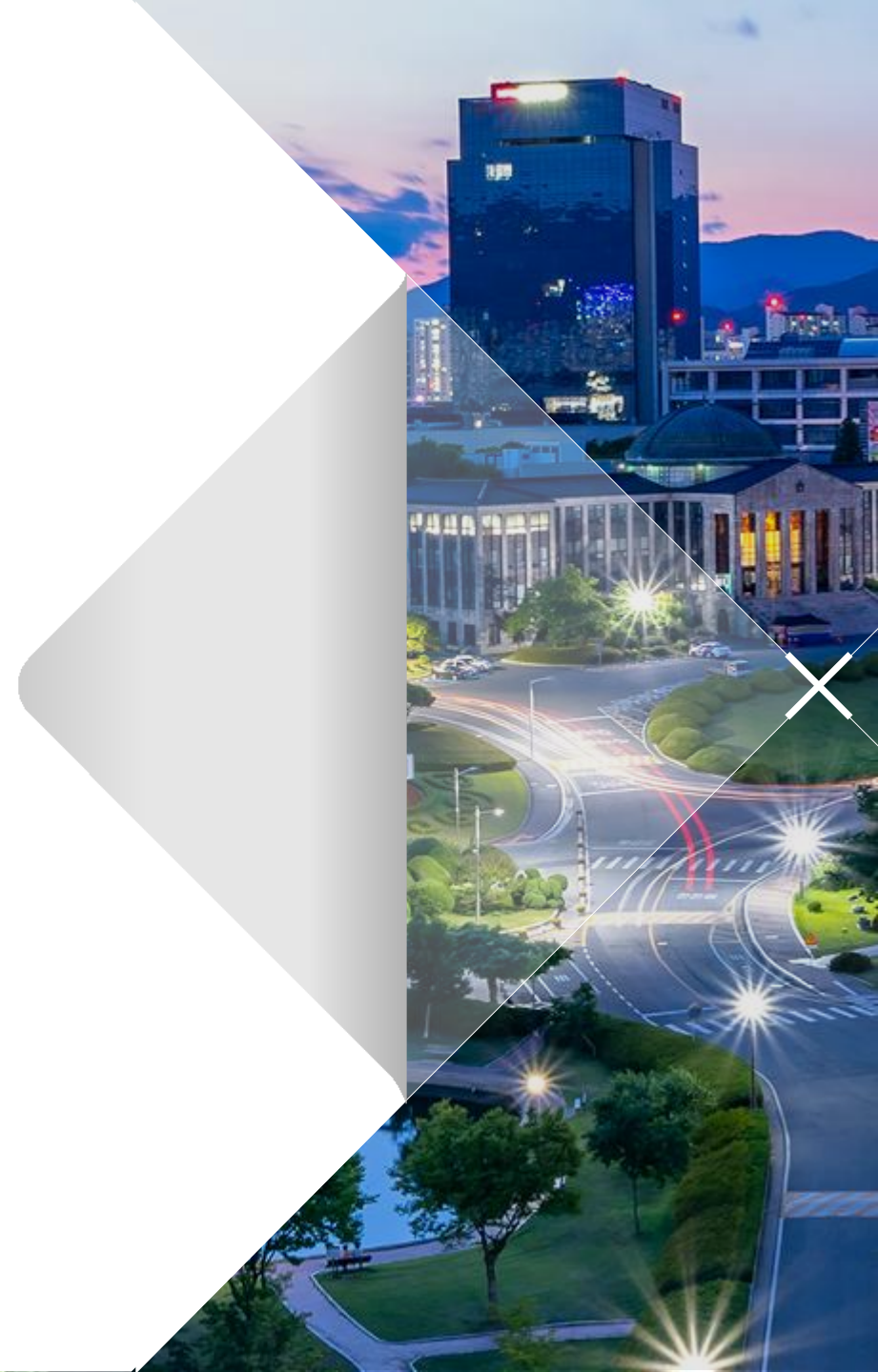
This message is shown once a day. To disable it please create the
/home/daegyeom/.hushlogin file.
(base) daegyeom@DESKTOP-RP6JG4F:~$
```

붙여넣기 : 우클릭

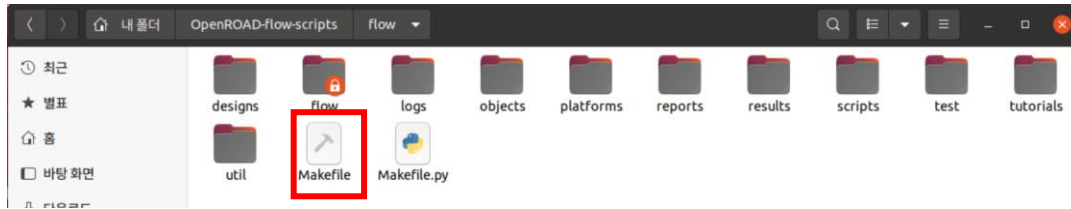
#반드시 경로 OpenROAD-flow-scripts/에서 실행

```
sudo docker run --rm -it -u $(id -u ${USER}):$(id -g ${USER}) -v
$(pwd)/flow:/OpenROAD-flow-scripts/flow openroad/flow-
ubuntu22.04-builder
source ./env.sh
cd flow
make DESIGN_CONFIG=./designs/asap7/aes/config.mk
```

III. Synthesis



For POST synthesis report



OpenROAD-flow-scripts/flow/Makefile

```
296 do-yosys-canonicalize: yosys-dependencies $(DONT_USE_SC_LIB)
297   $(SCRIPTS_DIR)/synth.sh $(SCRIPTS_DIR)/synth_canonicalize.tc
298
299 $(RESULTS_DIR)/1_synth.rtlil: $(YOSYS_DEPENDENCIES)
300   $(UNSET_AND_MAKE) do-yosys-canonicalize
301
302 $(RESULTS_DIR)/1_1_yosys.v: $(RESULTS_DIR)/1_synth.rtlil
303   $(UNSET_AND_MAKE) do-yosys
304
305 .PHONY: do-synth
306 do-synth:
307   mkdir -p $(RESULTS_DIR) $(LOG_DIR) $(REPORTS_DIR)
308   cp $(RESULTS_DIR)/1_1_yosys.v $(RESULTS_DIR)/1_synth.v
309
310 $(RESULTS_DIR)/1_synth.v: $(RESULTS_DIR)/1_1_yosys.v
311   $(UNSET_AND_MAKE) do-synth
312
```

```
296 do-yosys-canonicalize: yosys-dependencies $(DONT_USE_SC_LIB)
297   $(SCRIPTS_DIR)/synth.sh $(SCRIPTS_DIR)/synth_canonicalize.tc
298
299 $(RESULTS_DIR)/1_synth.rtlil: $(YOSYS_DEPENDENCIES)
300   $(UNSET_AND_MAKE) do-yosys-canonicalize
301
302 $(RESULTS_DIR)/1_1_yosys.v: $(RESULTS_DIR)/1_synth.rtlil
303   $(UNSET_AND_MAKE) do-yosys
304
305 .PHONY: do-synth
306 do-synth:
307   mkdir -p $(RESULTS_DIR) $(LOG_DIR) $(REPORTS_DIR)
308   cp $(RESULTS_DIR)/1_1_yosys.v $(RESULTS_DIR)/1_synth.v
309
310 $(RESULTS_DIR)/1_synth.v: $(RESULTS_DIR)/1_1_yosys.v
311   $(UNSET_AND_MAKE) do-synth
312   $(UNSET_AND_MAKE) do-synth-report
```

For report about Netlist after synthesis



~/OpenROAD-flow-scripts/flow/scripts/synth_canonicalize.tcl

```
source $::env(Scripts_DIR)/synth_preamble.tcl
read_design_sources

dict for {key value} $::env(VERILOG_TOP_PARAMS) {
    # Apply toplevel parameters
    chparam -set $key $value $::env(Design_NAME)
}

hierarchy -check -top $::env(Design_NAME)

set cmd "proc; opt;"
yosys $cmd
# Get rid of unused modules
opt_clean -purge
# The hash of this file will not change if files not part of synthesis do not change
write_rtlil $::env(Results_DIR)/1_synth.rtlil
```

```
22.2. Executing DEMUXMAP pass.
exec cp /home/son/OpenROAD-flow-scripts/flow/designs/asap7/chip/constraint.sdc ./results/asap7/chip/base/1_synth.sdc
End of script. Logfile hash: ed882ee7a5, CPU: user 170.68s system 1.20s, MEM: 3377.56 MB peak
Yosys 0.54 (git sha1 db72ec3bd, g++ 11.4.0-1ubuntu1~22.04 -fPIC -O3)
Time spent: 20% 2x abc (42 sec), 18% 1x extract_fa (39 sec), ...
Elapsed time: 3:30.50[h:]min:sec. CPU time: user 208.02 sys 2.35 (99%). Peak memory: 3458620KB.
mkdir -p ./results/asap7/chip/base ./logs/asap7/chip/base ./reports/asap7/chip/base
cp ./results/asap7/chip/base/1_1_yosys.v ./results/asap7/chip/base/1_synth.v
(env time -f 'Elapsed time: %E[h:]min:sec. CPU time: user %U sys %S (%P). Peak memory: %MKB.' /home/son/OpenROAD-flow-scripts/tools/install/OpenROAD/bin/op
>&1 | tee /home/son/OpenROAD-flow-scripts/flow/logs/asap7/chip/base/1_1_yosys_metrics.log
OpenROAD v2.0-22184-ga9002cfe20
Features included (+) or not (-): +GPU +GUI +Python
This program is licensed under the BSD-3 license. See the LICENSE file for details.
Components of this program may be licensed under more restrictive licenses which must be honored.
[INFO ORD-0030] Using 12 thread(s).
source /home/son/OpenROAD-flow-scripts/flow/platforms/asap7/liberty_suppressions.tcl
suppress_message STA 1212
[INFO ODB-0227] LEF file: /home/son/OpenROAD-flow-scripts/flow/platforms/asap7/lef/asap7_tech_1x_201209.lef, created 24 layers, 9 vias
[INFO ODB-0227] LEF file: /home/son/OpenROAD-flow-scripts/flow/platforms/asap7/lef/asap7sc7p5t_28_R_1x_220121a.lef, created 212 library cells
Report metrics stage 1, Post synthesis...

=====
Post synthesis report_design_area
=====
Design area 23100 u^2 100% utilization.
Elapsed time: 0:54.14[h:]min:sec. CPU time: user 62.25 sys 6.27 (126%). Peak memory: 1109744KB.
OpenROAD v2.0-22184-ga9002cfe20
Features included (+) or not (-): +GPU +GUI +Python
This program is licensed under the BSD-3 license. See the LICENSE file for details.
Components of this program may be licensed under more restrictive licenses which must be honored.
[INFO ORD-0030] Using 12 thread(s).
mkdir -p ./objects/asap7/chip/base
/home/son/OpenROAD-flow-scripts/flow/scripts/flow.sh 2_1_floorplan floorplan
Running floorplan.tcl, stage 2_1_floorplan
source /home/son/OpenROAD-flow-scripts/flow/platforms/asap7/liberty_suppressions.tcl
suppress_message STA 1212
[INFO ODB-0227] LEF file: /home/son/OpenROAD-flow-scripts/flow/platforms/asap7/lef/asap7_tech_1x_201209.lef, created 24 layers, 9 vias
[INFO ODB-0227] LEF file: /home/son/OpenROAD-flow-scripts/flow/platforms/asap7/lef/asap7sc7p5t_28_R_1x_220121a.lef, created 212 library cells

=====
Floorplan check_setup
=====
```

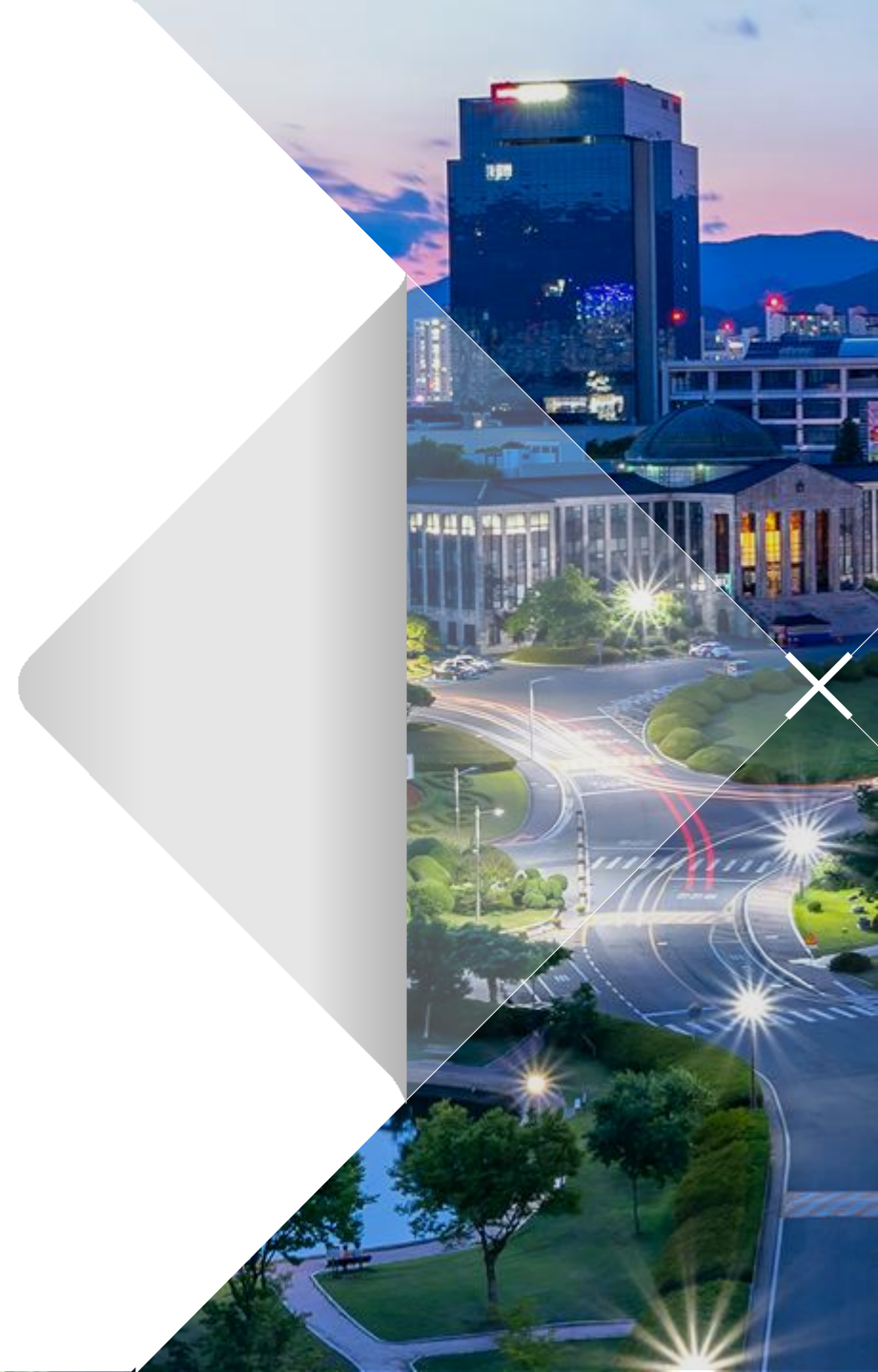
여기까지만 하고 ctrl + C 눌러서 종료하면 됨.
Floorplan 이후 과정은 진행할 필요 없음.

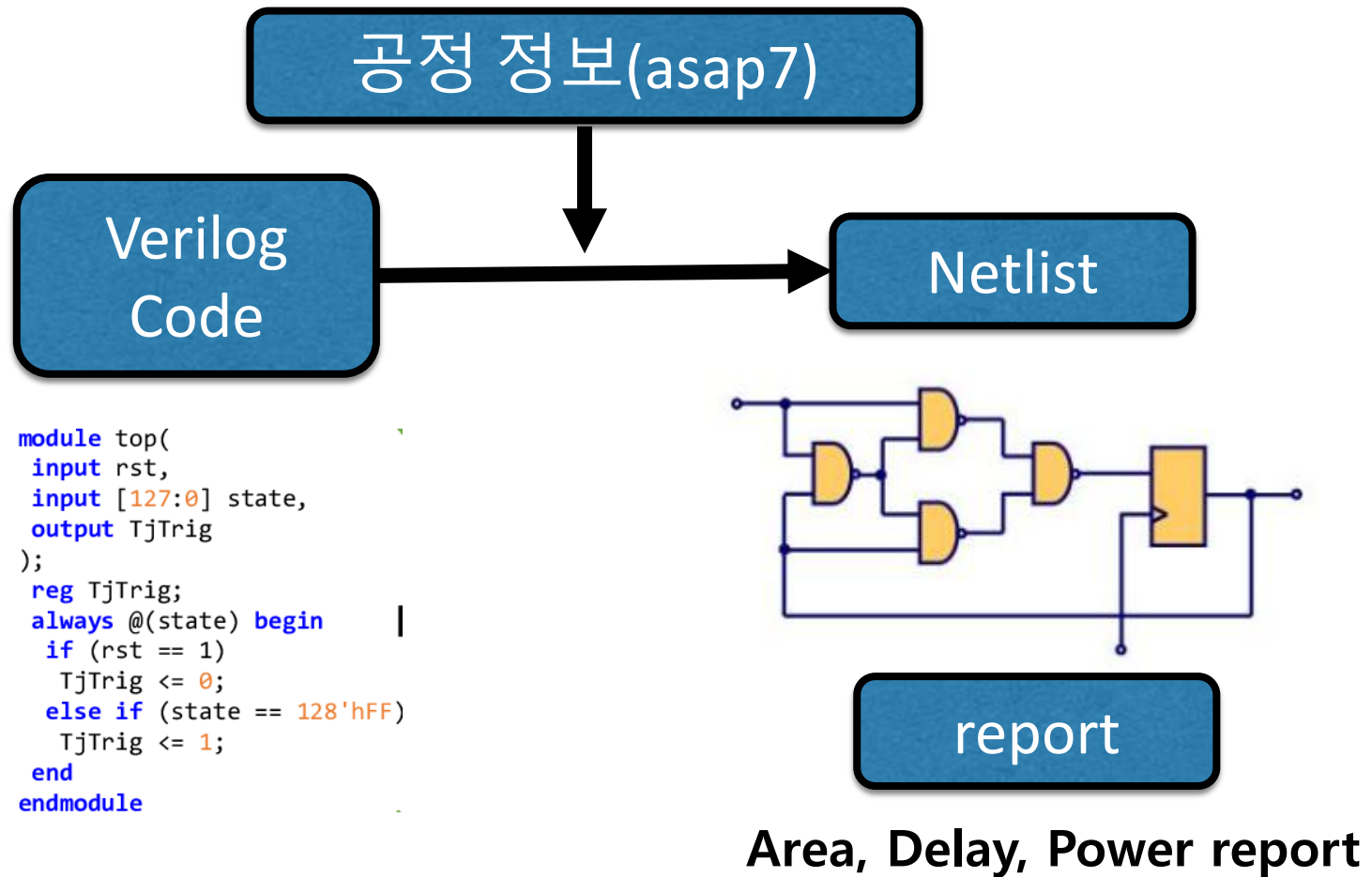
[illegible]

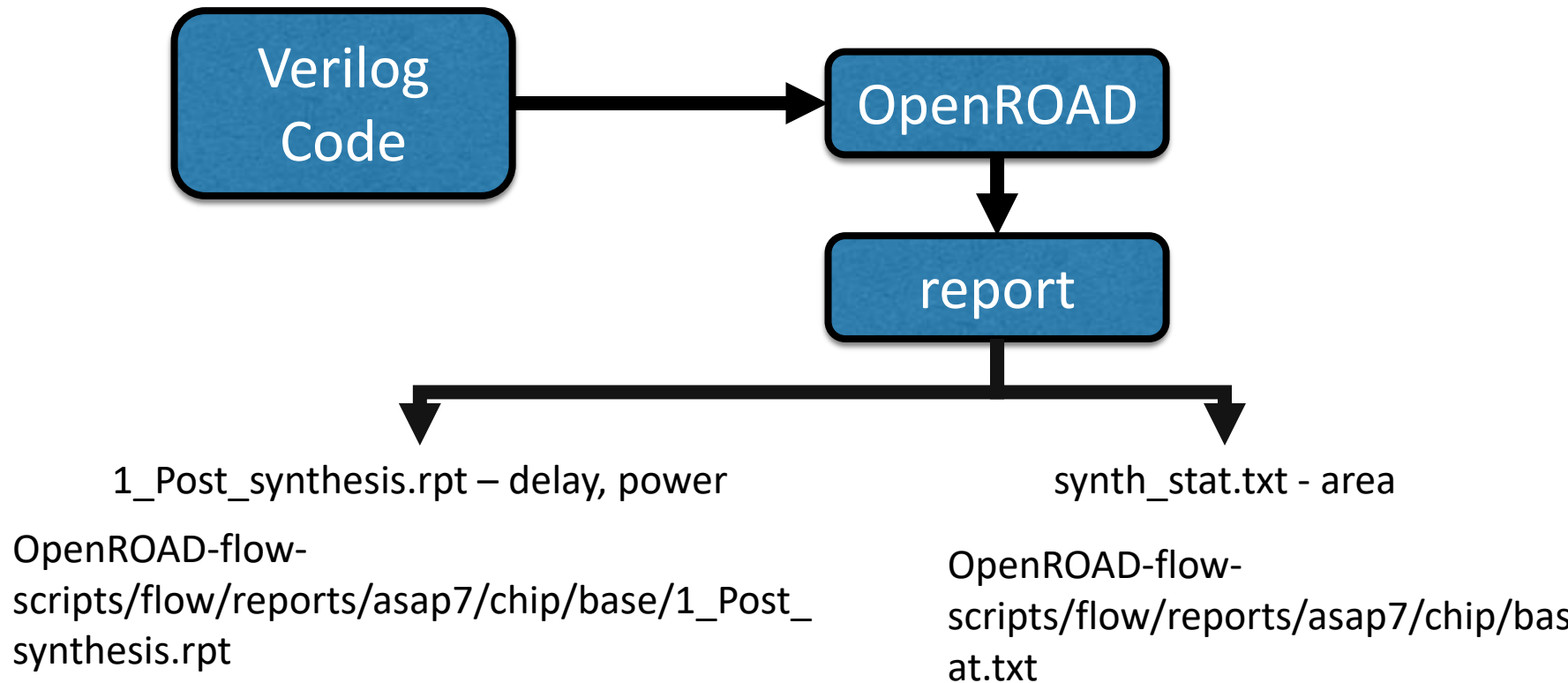
 asap7sc7p5t_AO_RVT_TT_201020	2025-06-23 오후 2:12	V 파일
 asap7sc7p5t_INVBUF_RVT_TT_201020	2025-06-23 오후 2:12	V 파일
 asap7sc7p5t_OA_RVT_TT_201020	2025-06-23 오후 2:12	V 파일
 asap7sc7p5t_SEQ_RVT_TT_220101	2025-06-23 오후 2:12	V 파일
 asap7sc7p5t_SIMPLE_RVT_TT_201020	2025-06-23 오후 2:12	V 파일
 empty	2025-06-23 오후 2:12	V 파일

제공한 파일 OpenRoad project/std cell/

IV. OpenROAD Flow







1_Post_synthesis.rpt – delay, power

```

=====
Post synthesis report_tns
-----
tns 0.00
=====
Post synthesis report_wns
-----
wns 0.00
=====
Post synthesis report_worst_slack
-----
worst slack 74.33
=====
Post synthesis report_checks -path_delay min
-----
Startpoint: conv1_layer.conv1_calc.latency[0]$_SDFF_PN0_
(rising edge-triggered flip-flop clocked by clk)
Endpoint: conv1_layer.conv1_calc.latency[1]$_SDFF_PN0_
(rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

Fanout    Cap    Slew    Delay    Time    Description
-----
1          0.78    12.99    32.45    32.45    clock clk (rise edge)
1          0.78    12.99    32.45    32.45    clock network delay (ideal)
1          0.78    12.99    32.45    32.45    ^ conv1_layer.conv1_calc.latency[0]$_SDFF_PN0_/CLK (DFFHQNx1_ASAP7_75t_R)
1          0.78    12.99    32.45    32.45    ^ conv1_layer.conv1_calc.latency[0]$_SDFF_PN0_/QN (DFFHQNx1_ASAP7_75t_R)
1          0.78    12.99    32.45    32.45    ^ conv1_layer._01544_ (net)
1          0.78    12.99    32.45    32.45    ^ conv1_layer._44514_/A (NOR2x1_ASAP7_75t_R)
1          0.78    12.99    32.45    32.45    ^ conv1_layer._44514_/Y (NOR2x1_ASAP7_75t_R)
1          0.78    12.99    32.45    32.45    ^ conv1_layer._04256_ (net)
1          0.78    12.99    32.45    32.45    v conv1_layer.conv1_calc.latency[1]$_SDFF_PN0_/D (DFFHQNx1_ASAP7_75t_R)
1          0.78    12.99    32.45    32.45    v data arrival time
1          0.78    12.99    32.45    32.45    clock clk (rise edge)
1          0.78    12.99    32.45    32.45    clock network delay (ideal)
1          0.78    12.99    32.45    32.45    clock reconvergence pessimism
1          0.78    12.99    32.45    32.45    ^ conv1_layer.conv1_calc.latency[1]$_SDFF_PN0_/CLK (DFFHQNx1_ASAP7_75t_R)
1          0.78    12.99    32.45    32.45    library hold time
1          0.78    12.99    32.45    32.45    data required time
1          0.78    12.99    32.45    32.45    data required time
1          0.78    12.99    32.45    32.45    data arrival time
1          0.78    12.99    32.45    32.45    33.66 slack (MET)

```

Slack 값이 음수이면, delay 스펙을 만족하지 못한다는 뜻임.

```

=====
Post synthesis report_power
-----
Group          Internal Power    Switching Power    Leakage Power    Total Power (Watts)
-----
Sequential      2.34e-02          6.43e-04          2.15e-06          2.40e-02          35.7%
Combinational    1.68e-02          2.64e-02          2.50e-05          4.32e-02          64.3%
Clock            0.00e+00          0.00e+00          0.00e+00          0.00e+00          0.0%
Macro           0.00e+00          0.00e+00          0.00e+00          0.00e+00          0.0%
Pad              0.00e+00          0.00e+00          0.00e+00          0.00e+00          0.0%
-----
Total            4.02e-02          2.70e-02          2.72e-05          6.72e-02          100.0%
                    59.8%            40.2%            0.0%

```

Slack 값이 음수인 경우에는
power가 제대로 측정되지 않음.



synth_stat.txt - area

25. Printing statistics.

=== chip ===

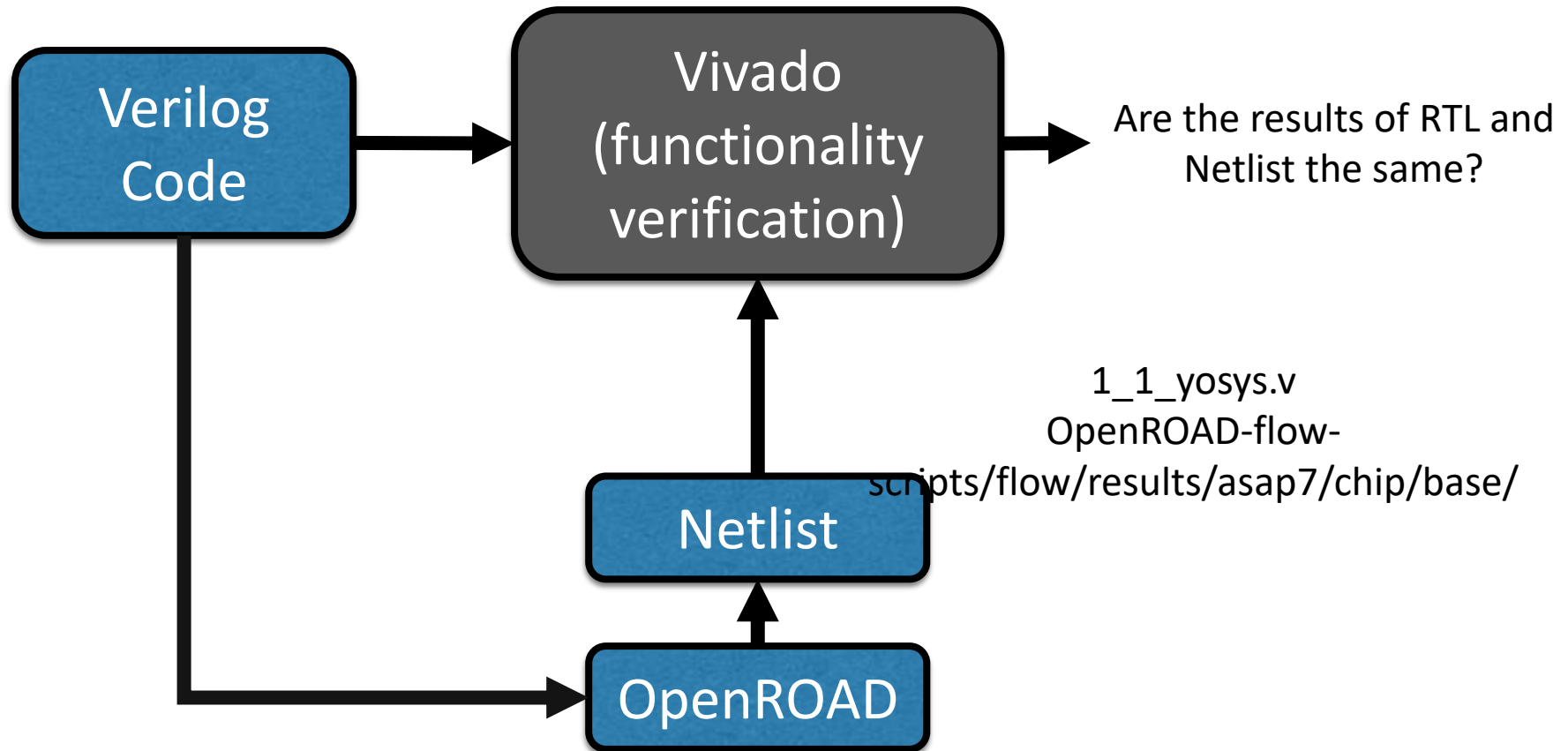
Number of wires:	303214
Number of wire bits:	307304
Number of public wires:	303214
Number of public wire bits:	307304
Number of ports:	39
Number of port bits:	4129
Number of memories:	0
Number of memory bits:	0
Number of processes:	0
Number of cells:	291464
AND2x2_ASAP7_75t_R	13730
AND2x4_ASAP7_75t_R	270
....	
XOR2x1_ASAP7_75t_R	2317
XOR2x2_ASAP7_75t_R	5022
Chip area for module '\chip'	33117.507720

of which used for sequential elements: 4045.950000 (12.22%)



Vivado

- Netlist와 stdcell Verilog file을 Vivado에 testbench와 함께 추가한 뒤, 시뮬레이션 진행
- Synthesis 이전과 같은 결과인지 확인

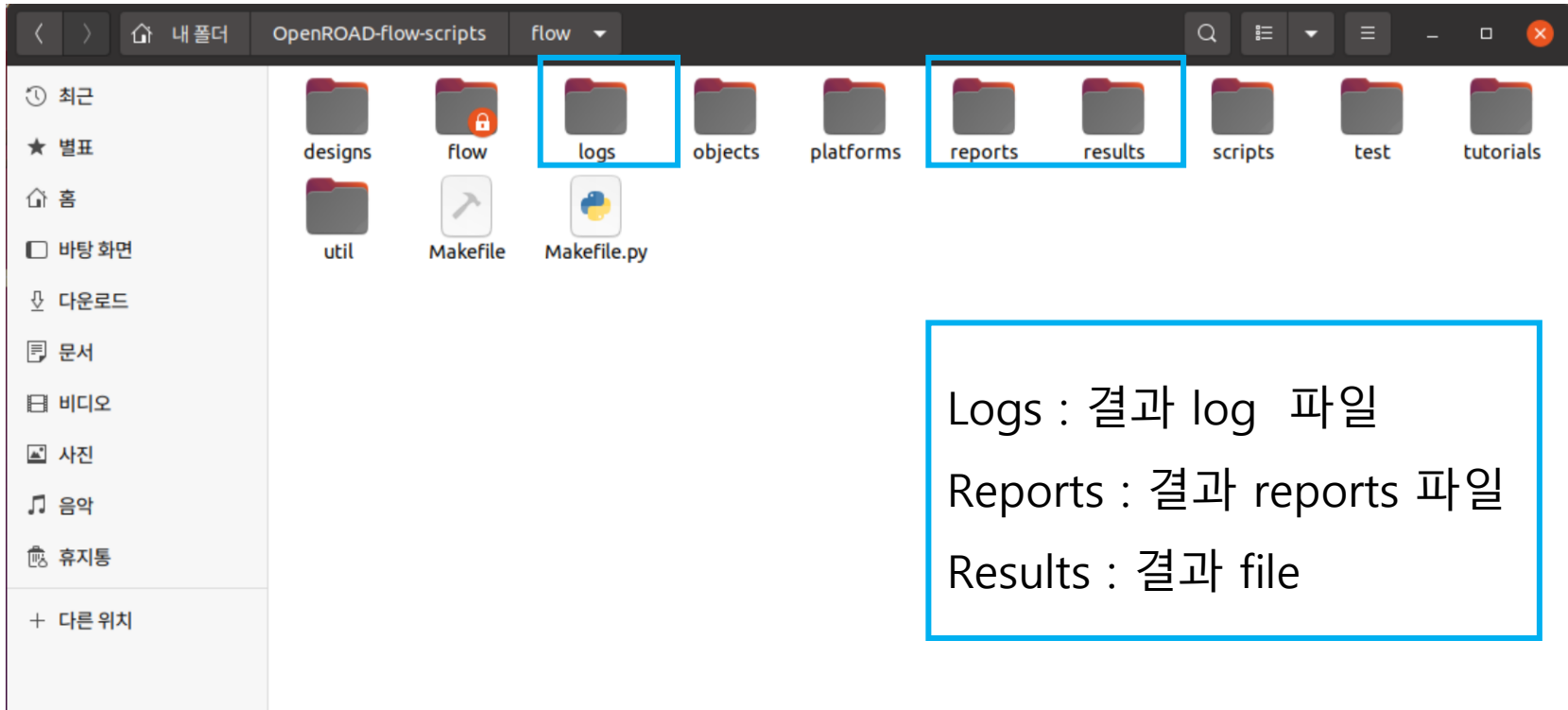


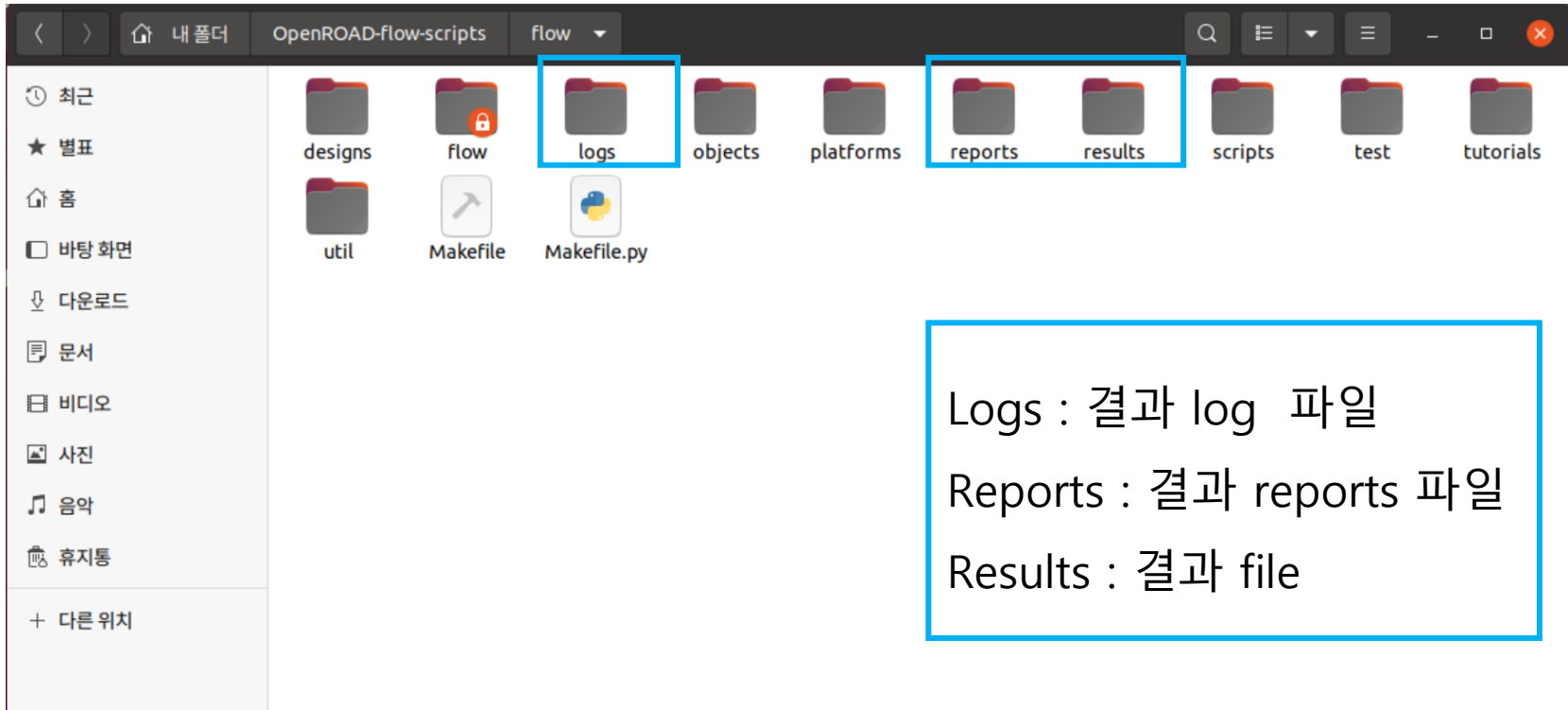
Verilog file 저장 위치

OPENROAD Project에서 제공하는 공정

OpenROAD 실행 file 위치

실행 파일





Project - functionality



Cloud > 2. Class > 논리회로설계 > openroad_project >

정렬 보기

이름	상태	수정한 날짜	유형	크기
config		2025-06-21 오후 2:54	파일 폴더	
std_cell		2025-06-21 오후 2:45	파일 폴더	
synth result		2025-06-21 오후 2:50	파일 폴더	
verilog		2025-06-21 오후 2:44	파일 폴더	

← Std cell code

Vivado 2024.1@iceku-srv1
File Flow Tools Window Help Q-Quick Access

AMD
Vivado
ML Edition

Netlist + testbench+ stdcell 모두 Add file

Quick Start

- Create Project >
- Open Project >
- Open Example Project >

Tasks

- Manage IP >
- Open Hardware Manager >
- Vivado Store >

Learning Center

- Documentation and Tutorials >
- Quick Take Videos >
- What's New in 2024.1 >

Recent Projects

- MIPS_CPU
/home/XNUICEHdd1/JCE/seunghyun/MIPS_CPU
- 0314
/home/XNUICEHdd1/JCE/seunghyun/0314

Add Source Files@iceku-srv1

Look in: final_project_logic_design

- 1_1_yosys.v
- asap7sc6t_INVBUF_RVT_FF_210930.v
- asap7sc7p5t_AO_RVT_TT_201020.v
- asap7sc7p5t_INVBUF_RVT_TT_201020.v
- asap7sc7p5t_INVBUF_RVT_TT_201020.v
- asap7sc7p5t_OA_RVT_TT_201020.v
- asap7sc7p5t_SEQ_RVT_TT_220101.v
- asap7sc7p5t_SIMPLE_RVT_TT_201020.v

Recent Directories

- /home/XNUICEHdd1/JCE/seunghyun

File Preview

File: 1_1_yosys.v
Directory: /home/XNUICEHdd1/JCE/seunghyun/Vivado/final_project_logic_de
Created: Today at 16:09 PM
Accessed: Today at 16:09 PM
Modified: Today at 16:09 PM
Size: 57.5 MB
Type: Netlist description
Owner: seunghyun
Permissions: rw-r--r--

File name: "asap7sc7p5t_AO_RVT_TT_201020.v" "asap7sc7p5t_INVBUF_RVT_TT_201020.v" "asap7sc7p5t_OA_RVT_TT_201020.v" "asap7sc7p5t_SEQ_RVT_TT_220101.v" "asap7sc7p5t_SIMPLE_RVT_TT_201020.v"
Files of type: Design Source Files (.vhd, .vhdl, .vhdp, .vho, .v, .vf, .verilog, .v, .vg, .vb, .tf, .vlog, .vp, .vm, .veo, .svo, .vh, .h, .svh, .vhp, .svhp, .edn, .edf, .ngc, .sv, .snp, .bmm, .mif, .mem, .elf, .dcp, .bd, .wcfg, .xco, .xci, .xcix)

OK Cancel

The screenshot displays the Vivado IDE interface during a behavioral simulation of the `top_tb_1000` project. The **Flow Navigator** on the left shows the project structure, including the **PROJECT MANAGER**, **IP INTEGRATOR**, **SIMULATION**, **RTL ANALYSIS**, **SYNTHESIS**, and **IMPLEMENTATION** tabs. The **SIMULATION** tab is active, showing the **Run Simulation** button.

The **Scope** window displays the design hierarchy:

Name	Design Unit	Block Type
top_tb_1000	top_tb_1000	Verilog Moc
chip1	chip	Verilog Moc
gbl	gbl	Verilog Moc

The **Objects** window shows the current values of various signals:

Name	Value
clk	X
rst_n	X
pixels[0:783999][7:0]	XX,XX,XX,XX,XX
img_idx[9:0]	XXX
data_in[7:0]	XX
cnt[9:0]	XXX
input_cnt[9:0]	XXX
rand_num[9:0]	XXX
state	X
i_cnt[31:0]	X
accuracy[9:0]	XXX
decision[3:0]	X
valid_out_6	X
weight_11[0:24][7:0]	XX,XX,XX,XX,XX
weight_12[0:24][7:0]	XX,XX,XX,XX,XX

The **Waveform** window shows the timing diagram for the signals. The time scale is 165,660 ps. The signals shown include `clk`, `rst_n`, `img_idx[9:0]`, `data_in[7:0]`, `cnt[9:0]`, `input_cnt[9:0]`, `rand_num[9:0]`, `state`, `i_cnt[31:0]`, `accuracy[9:0]`, `decision[3:0]`, and `valid_out_6`.

The **Tcl Console** at the bottom shows the following messages:

```
9th input image : original value = 9, dec
10th input image : original value = 5, de
11st input image : original value = 6, de
```

A **Run Simulation** dialog box is open in the foreground, showing the current time as 165675 ps. The dialog has a progress bar and buttons for **Background** and **Cancel**.

Top_tb_1000 을 실행하면 accuracy 확인 및 functionality 확인 가능

Thank you

